GEO INFORMATION TECHNOLOGY IN DIASTER MANAGEMENT IN SRILANKA

SP2409 - Geo-Spatial Analysis

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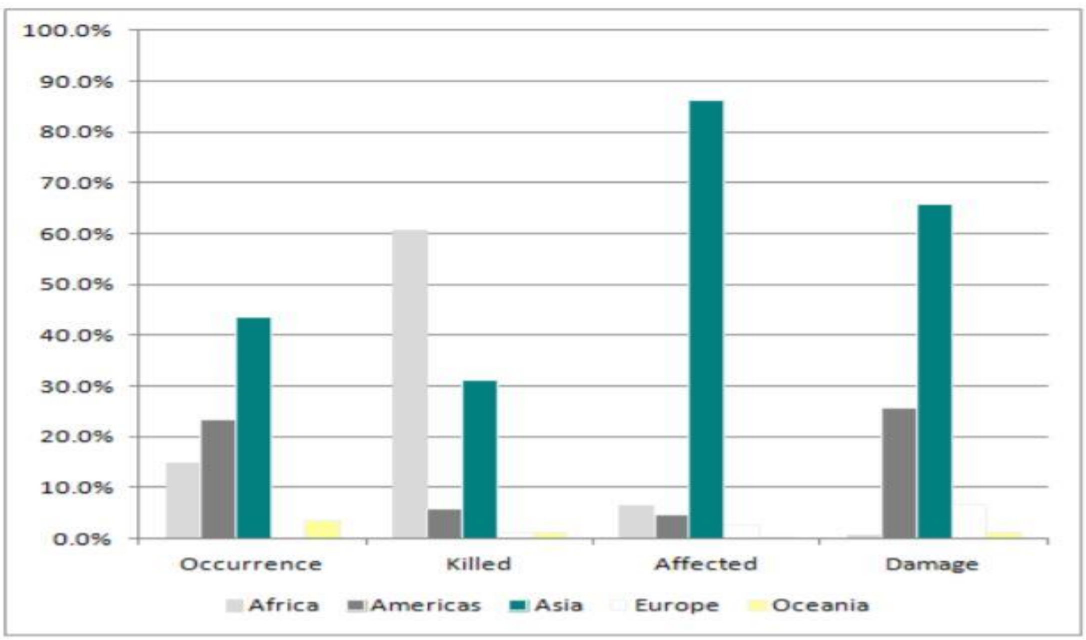
Statement of the Problem

Sri Lanka is located in the prone belt of the Asian Region. Based on world disaster records, Asia is identified as most affected region disaster. Accordingly, 48% of disasters happened in Asia.85 % exaggerated and 86% damaged globally by disaster in Asia.

Disasters by Continent (2005 – 2014)

Source: World Disaster report 2014



Source: World Disaster Report 2014

Therefore, Sri Lanka also affected by disasters. Based on the records, Eastern and western region of Sri Lanka are affected by water- related disasters. Like cyclone, drought, diseases, flooding, storm flows and tsunami.

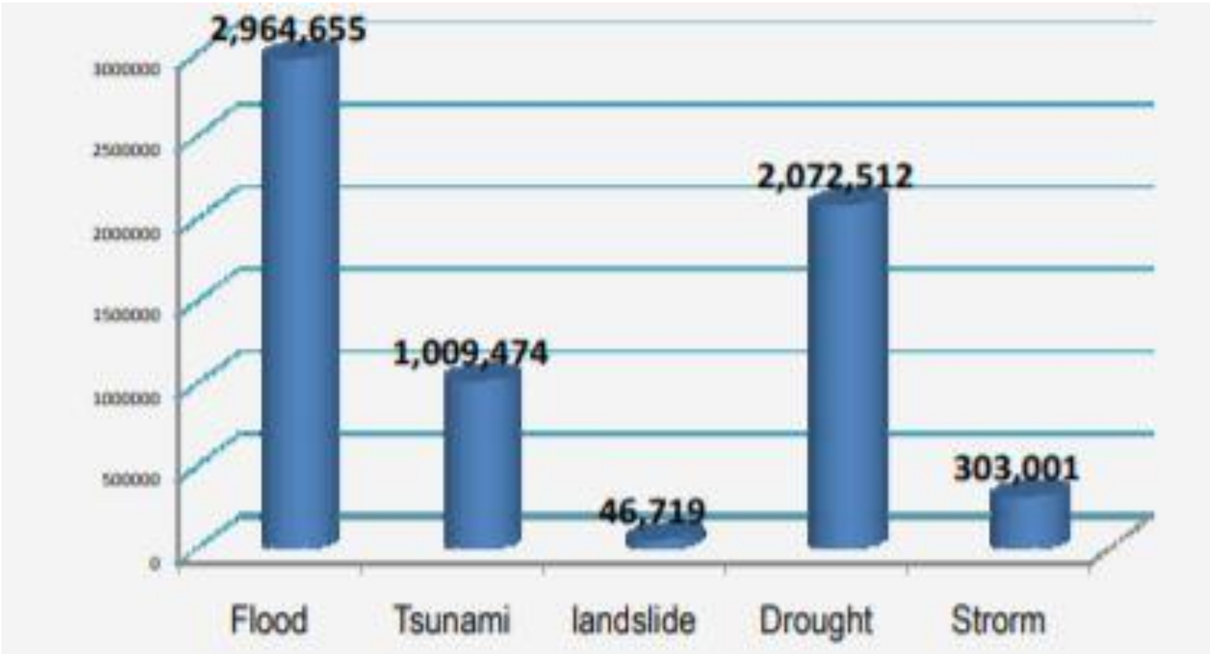
Disaster Profile of Sri Lanka

Disaster in Sri Lanka

Disaster	Year	Area	Dead	Affected
Flood	2000	Galle, Matara	02	100000
Flood	2000	Ampara, Batticaloa, Polonnaruwa	03	300000
Cyclone	2000	Ampara, Anuradhapura, Batticaloa, Mannar, Trincomalee, Polonnaruwa	05	375000
Flood	2001	Mattale		375000
Flood	2002	Ampara, Anuradhapura, Batticaloa, Mannar, Trincomalee, Polonnaruwa, Puttalam, Kilinochchi	02	500000
Flood	2003	Galle, Matara, Hambantota, Nuwara Eliya, Kaluthra	296	695000
Flood	2004	Ampara, Anuradhapura, Batticaloa, Mannar, Trincomalee, Polonnaruwa	06	200000
Tsunami	2004	Jaffna, Mullaithivu, Kilinochchi, Ampara, Galle, Matara, Hambantota, Batticaloa	35399	23176
Flood	2005	Colombo, Ratmalana, Gamphaha, Puttalam, Matara, Badulla, Ratnapura	06	145000
Flood	2006	Colombo, Ratnapura, Gamphaha, Puttalam, Matara, Badulla, Ratmalana	25	333000
Flood	2007	Walappana, Meepai	18	68281

Source: Department of Disaster Management

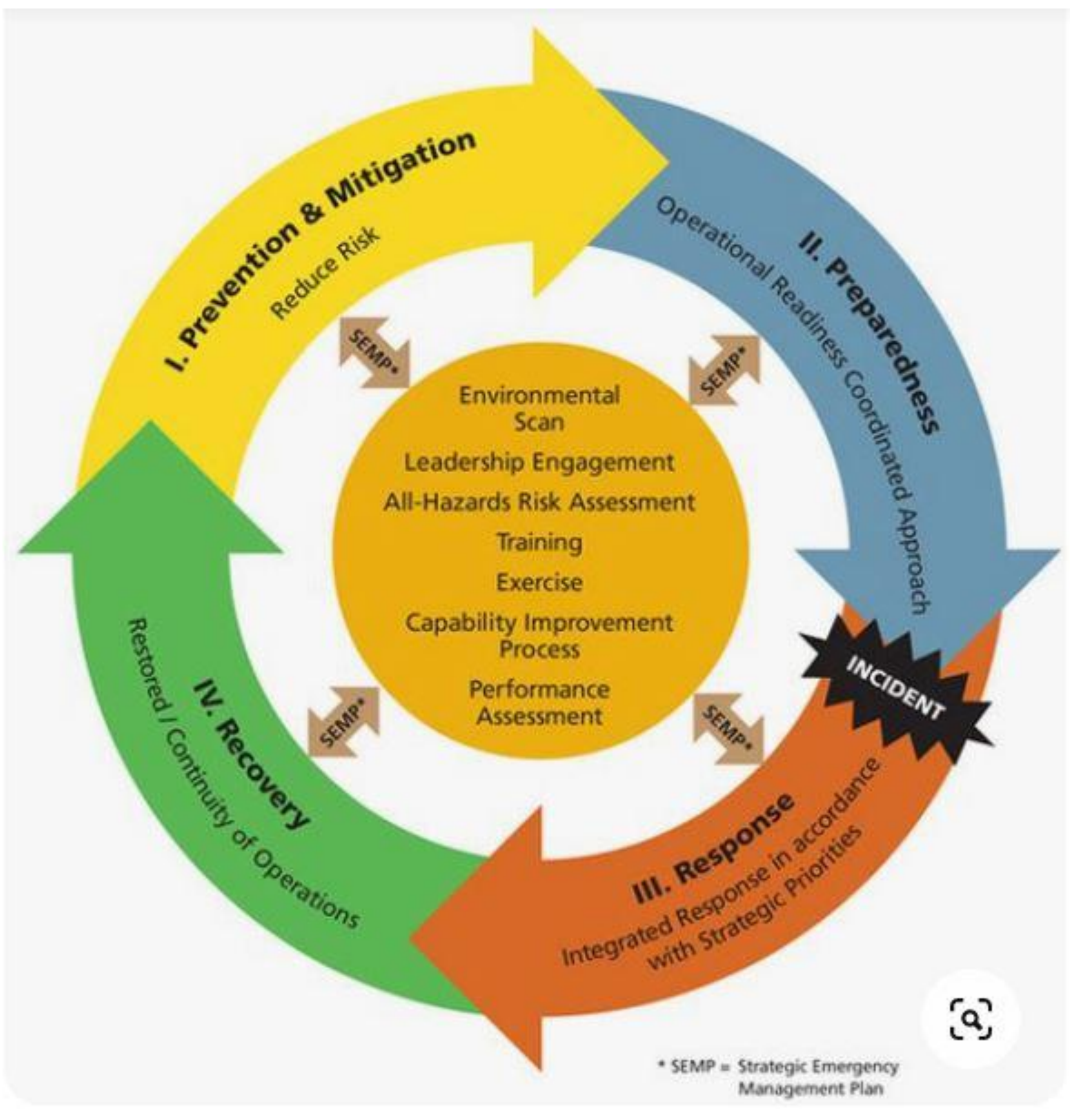
Number of People affected by disasters in Sri Lanka (2000-2008)



Natural and human disaster affected physical, social and economic development of Sri Lanka. Based on the records, Sri Lanka highly affected by Flood disaster. Accordingly, I consider flood for study.

Description of Disaster Management Circle

Natural disasters are unavoidable and it is practically difficult to fully recover from the damage affected by disasters. Accordingly consider the Disaster Management Circle, the goal is to reduce or prevent damage caused by natural disasters, provide timely and adequate assistance to victims of natural disasters, and recover quickly and effectively. The disaster management cycle describes the ongoing process of government, business, and civil society to mitigate the impact of a disaster, respond immediately during and after a disaster, and take action to recover from a disaster. Taking appropriate action at all stages of the cycle can improve prevention capacities in the subsequent iteration of the cycle, improve early warning, reduce vulnerability, or prevent disasters. The complete disaster management cycle includes the development of government policies and plans to eliminate the cause of the disaster or reduce its impact on personnel, property, and infrastructure.



Based on Disaster Management Circle, Mitigation is phase selected for analysis purposes

Mitigation

Disaster mitigation activities eliminate or reduce the chances of disasters or reduce the impact of unavoidable disasters. Building regulations, mitigation measures mitigation analysis updates; Zoning and territorial planning; Building use rules and safety rules; Preventive health care; And public education.

Relief depends on the inclusion of appropriate measures in national and regional development plans. Its effectiveness also depends on the availability of information on accidents, emergencies and adverse actions to take. The mitigation phase and even the entire disaster management cycle includes the development of public policies and plans to change the cause of the disaster or reduce the impact of the disaster on personnel, property and infrastructure.

How to Geo Informatics tools used for Disaster Management Circle?

Geo information tools are helped to analysis the large datas of ground and understand the condition of ground. It helps to identify the reasons of disaster, prevention, Preparation for safety and relief.

1. Watershed or River Basin Management
2. Disaster Assessment
3. Emergency planning and management of Disaster
4. Flood Risk Assessment
5. Land use management and planning

Current Situation Floods in Sri Lanka.

A flood is an unusual high stage in river or sea – normally the level at which the river/sea overflows its banks and inundates the adjoining area.

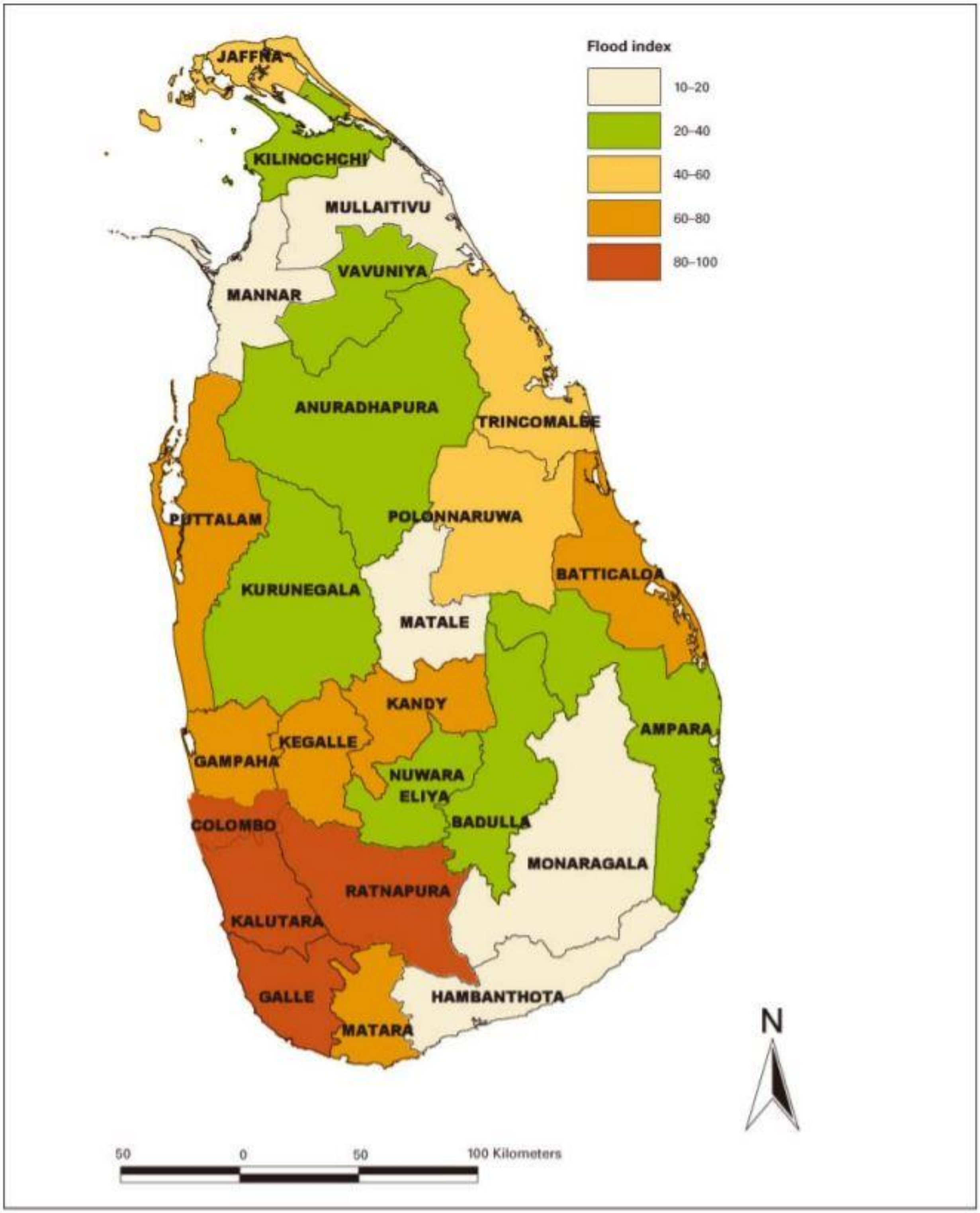
Classification of Floods in Sri Lanka

In Sri Lanka, Flood is classified based on causes and nature of floods.

- Riverine Floods
- Flash Floods
- Localized Floods
- Floods because of reservoirs
- Floods because of reservoirs breaking

Type of Floods	Causative Factors
Riverine Floods	• Overflowing of rivers •
Flash Floods	• Excessive rainfalls
Localized Floods	• Urbanization process
Floods because of reservoirs	• The operation of spillway gates of a reservoir
Floods because of reservoirs breaking	• Faulty design of reservoir • Defective constructions • Lack of proper maintenance

Flood hazard frequency map prepared by summarizing the number of floods registered in each district in the period 2000-2007.

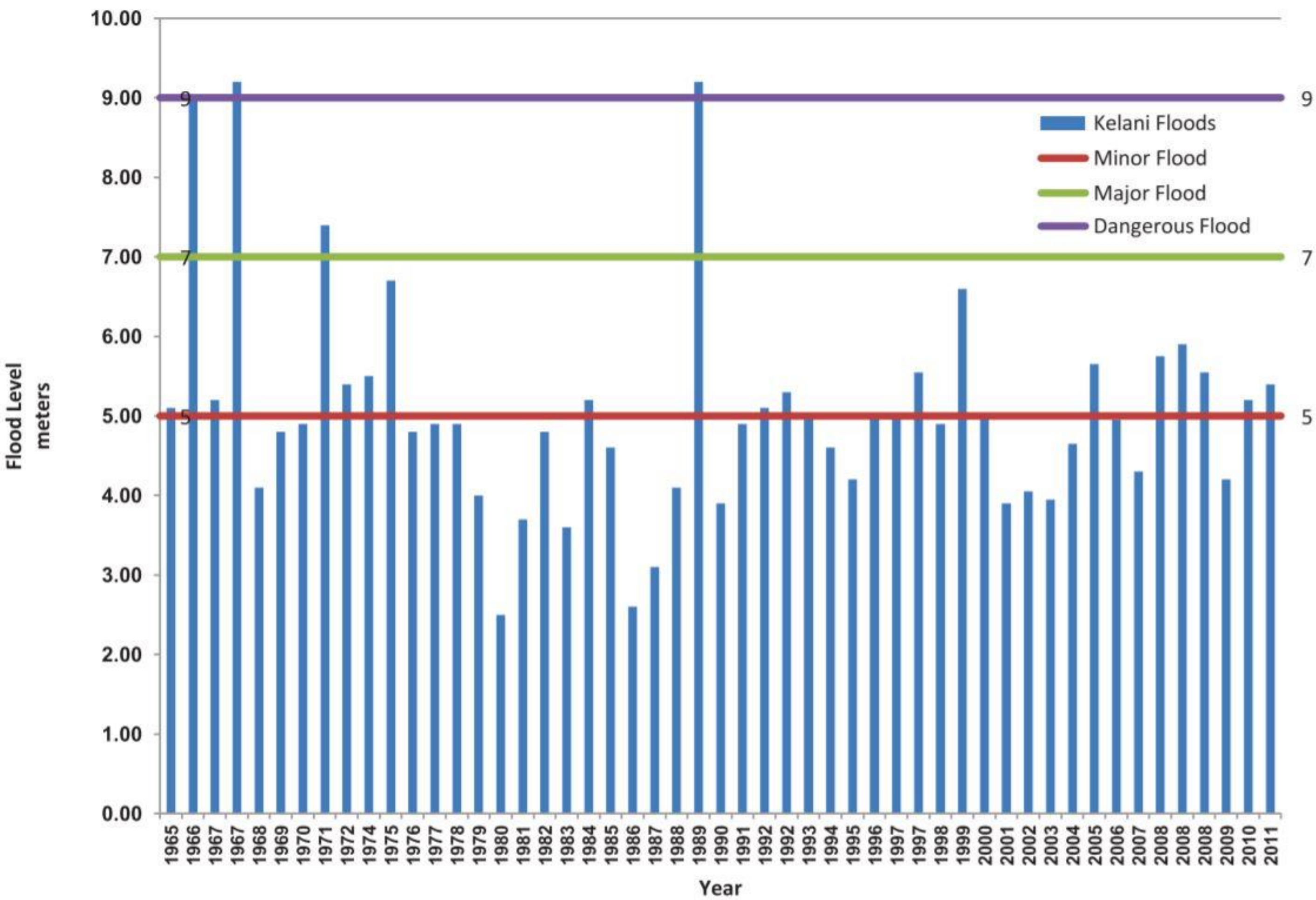


The flood Hazard map show high risk districts in Sri Lanka. Western province is identified as high risk zone in Sri Lanka. When considering the northern and eastern of country is affected by seasonal flood risks because of heavy rainfalls. In some areas, drainage, topography, and land use are also important factors. Floods affect people, economic activities and infrastructure. High-risk areas on the western Province have a high population density, a high concentration of industrial activity and infrastructure, and a high GDP.

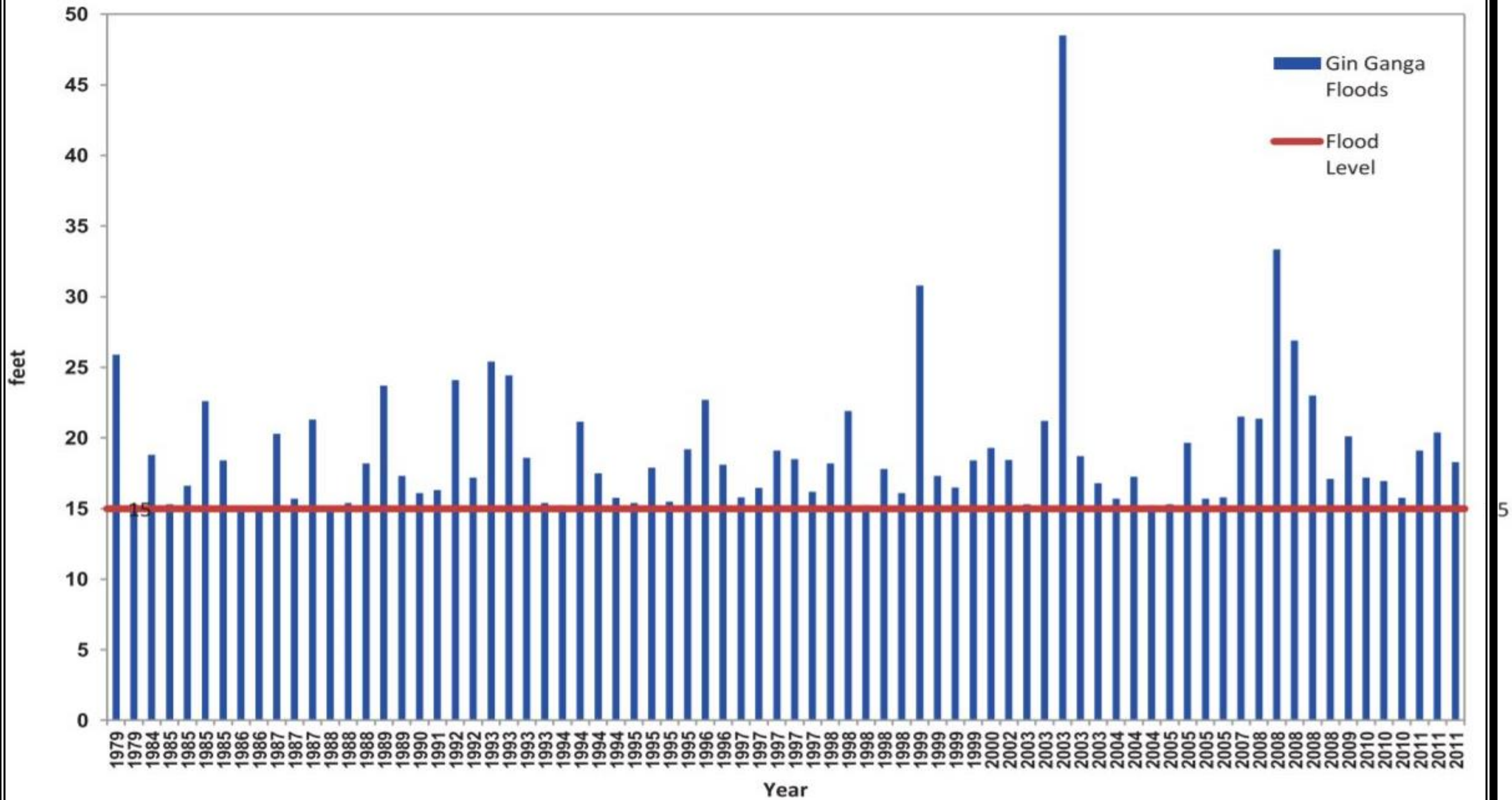
Statistics of Flood disaster occurrence in Sri Lanka.

Statistics of flood occurrences in some main rivers in Sri Lanka. The datas collected from Department of Irrigation. The datas are graphically explain the outline of flood in main rivers in Sri Lanka.

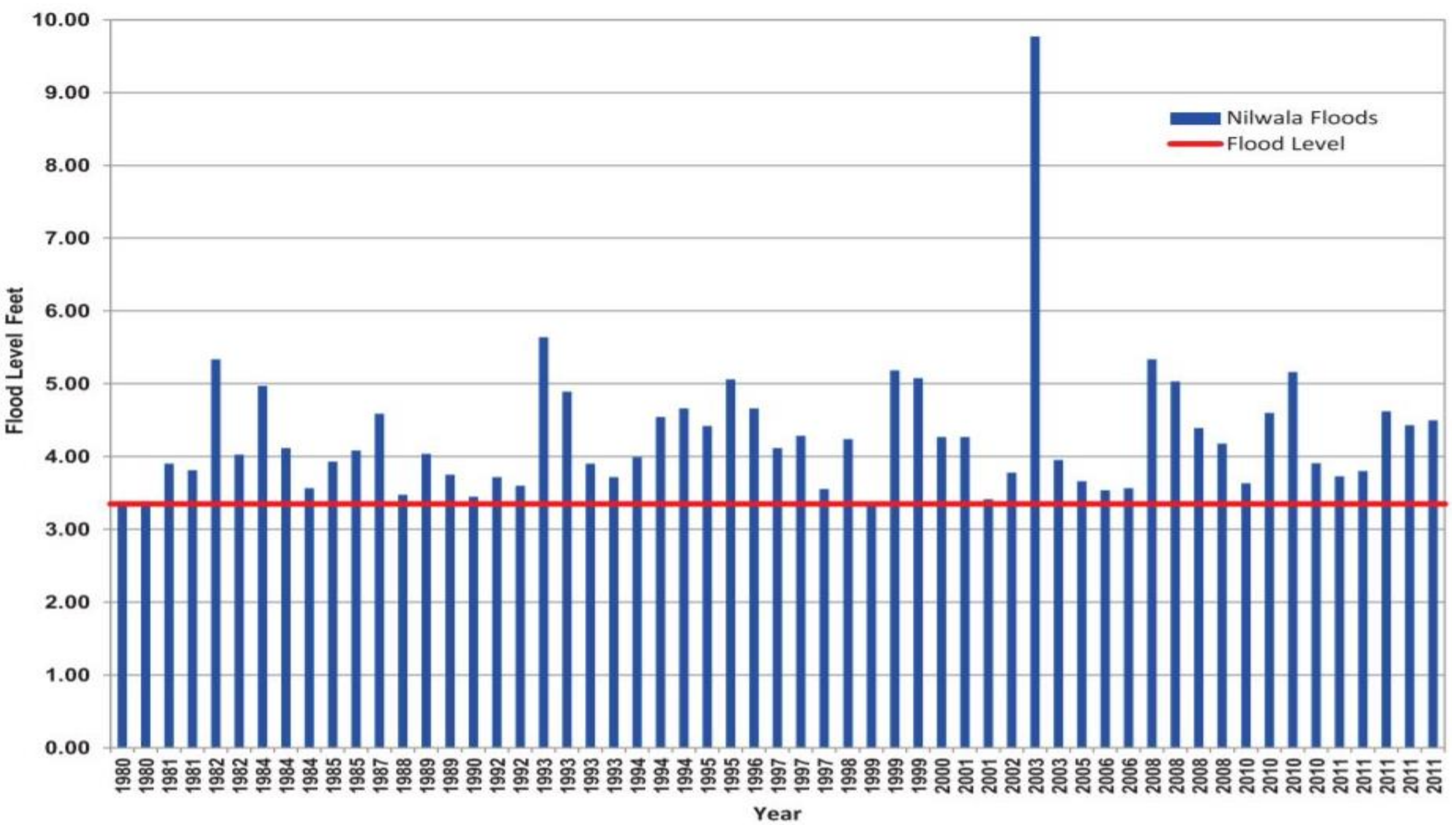
Kelani River Floods



Gin Ganga Floods



Nilwala Floods



Methodology

1. Spatial data inventory for the flood in Sri Lanka

Primary Data collection is not possible because of corona pandemic situation. Therefore, I used the Secondary Data Sources for the study.

Secondary Data Collection

Secondary data were got from published and unpublished sources. Initial information and mapping of geology and land use was carried out from information obtained by the Surveyors Office in Colombo, Sri Lanka.

Data	Sources	Remarks
SL DEM	TCP Common Server	Projection – GCS WGS 1984
Soil data	TCP Common Server	Projection – SLD 99 Re project – GCS WGS 1984
Contours data	Google earth	Using by SL DEM
Places and Location	OSM sever	Shapefiles are downloaded from UGCS Earth Explorer. Projection – GCS WGS 1984
Population data	TCP Common Server	Projection – SLD 99 Re project – GCS WGS 1984
GN Division Data	TCP Common Server	Projection – SLD 99 Re project – GCS WGS 1984
Building layer	RS Images	Last updated 2012 Therefore, manually updated with help of RS images and UGCS earth explorer.
River gauges and weather station	Central Environmental Authority	Digitalizing Using by Arc map 10.4
Flood based datas	Central Environmental Authority Ministry of Disaster Management Surveyor Department Disaster Management Centre of Sri Lanka (http://www.dmc.gov.lk/) Disaster Management Centre of Sri Lanka (http://www.dmc.gov.lk/) Department of Meteorology (http://www.meteo.gov.lk/) National Disaster Relief Services Centre (http://www.ndrsc.gov.lk/)	Secondary datas are used for analysis of the Study.

2. Approaches and Tools

I used QGIS Desktop 2.18.15 with GRASS 7.2.2 to water delineations of Kelani River Basin and Arc GIS 10.4 is used for model builder.

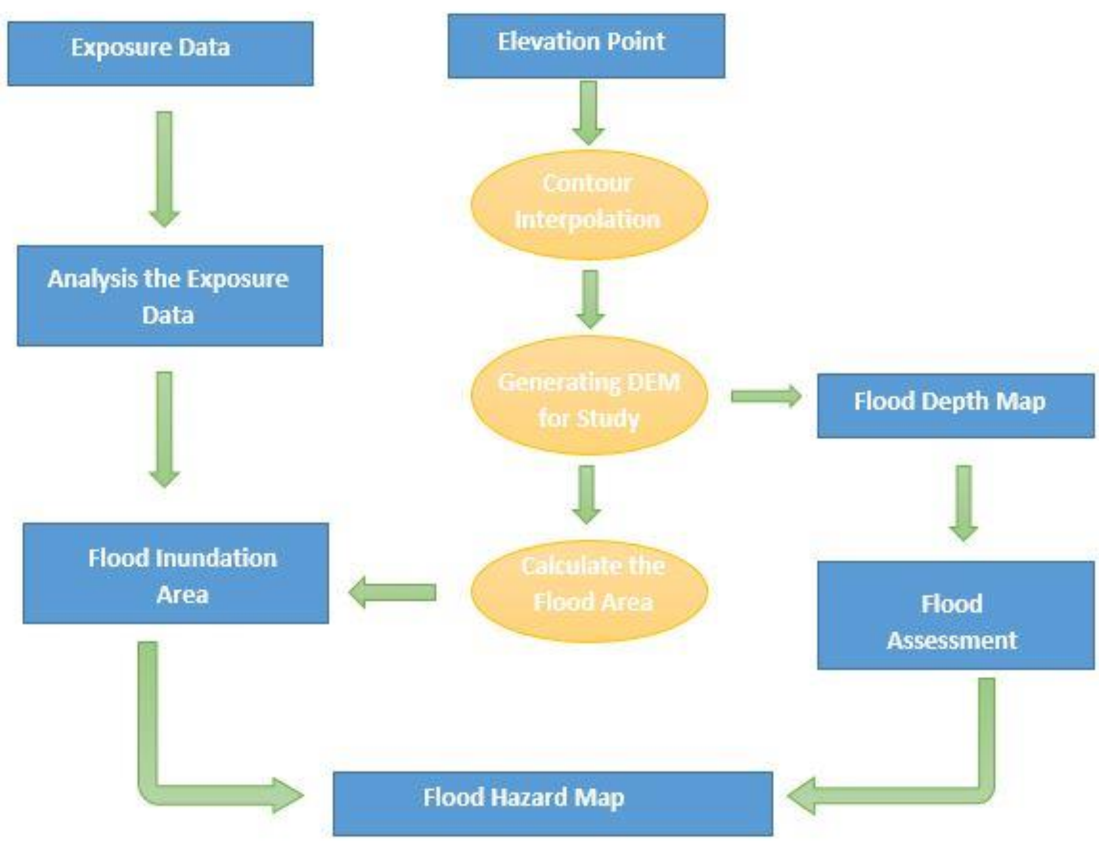
Use the UGCS earth explore as a source for collecting ground data of Kelani River Basin.

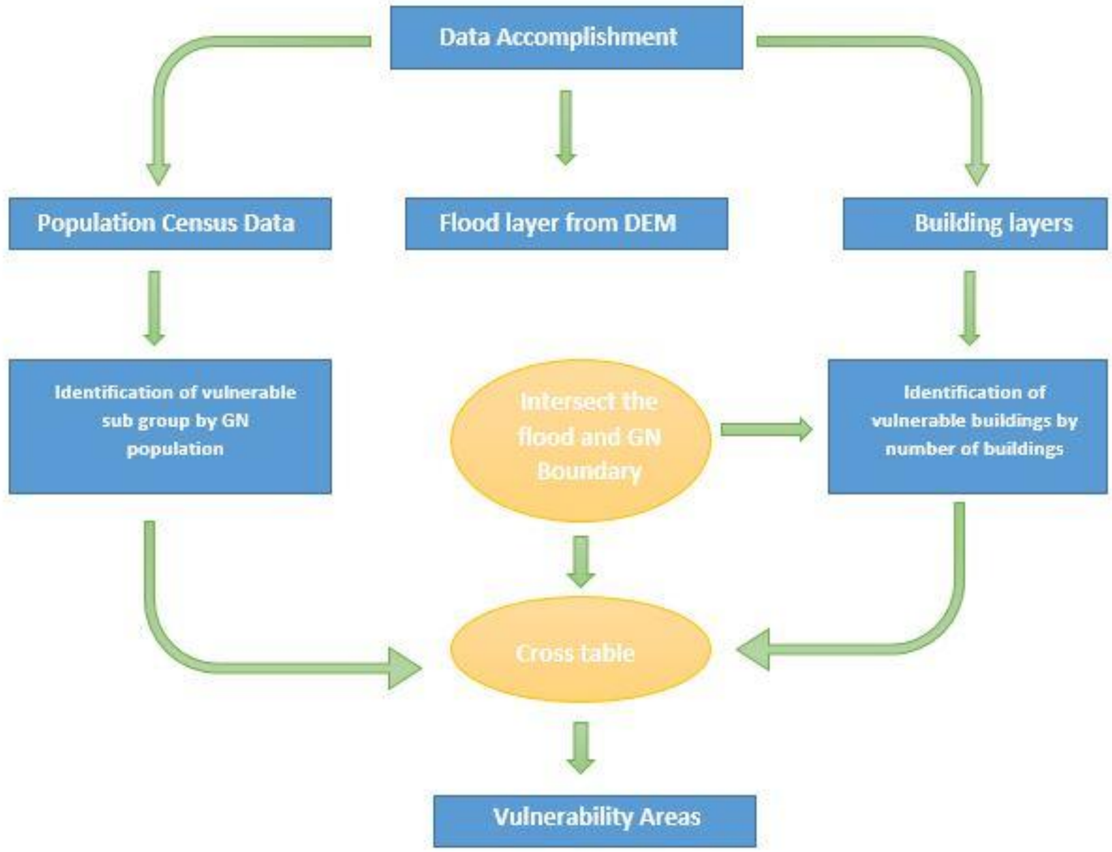
1. DEM of Sri Lanka
2. Building footprints
3. Roads
4. Places

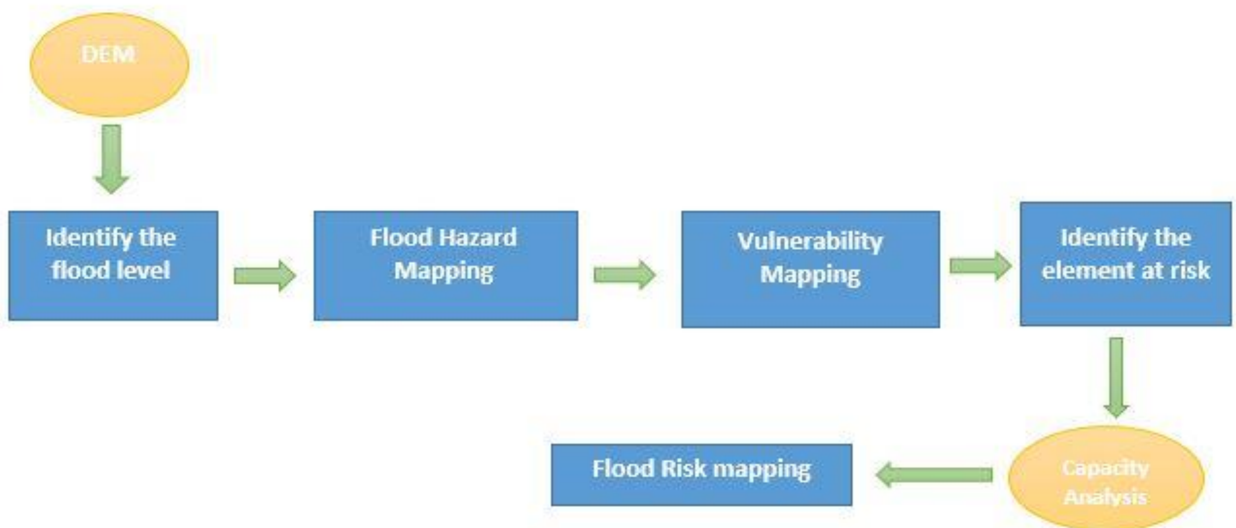
UGCS earth is the tool for downloading shapefiles from Open Street Map server.

Using Sri Lankan DEM to create a contour map.

Generate Flood, Vulnerability and Risk Maps of Kelani River Basin

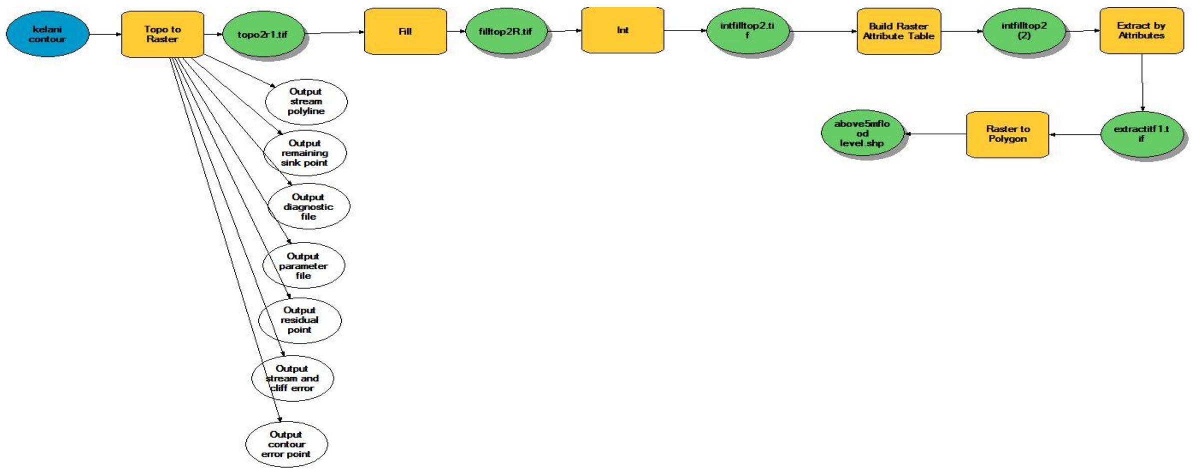


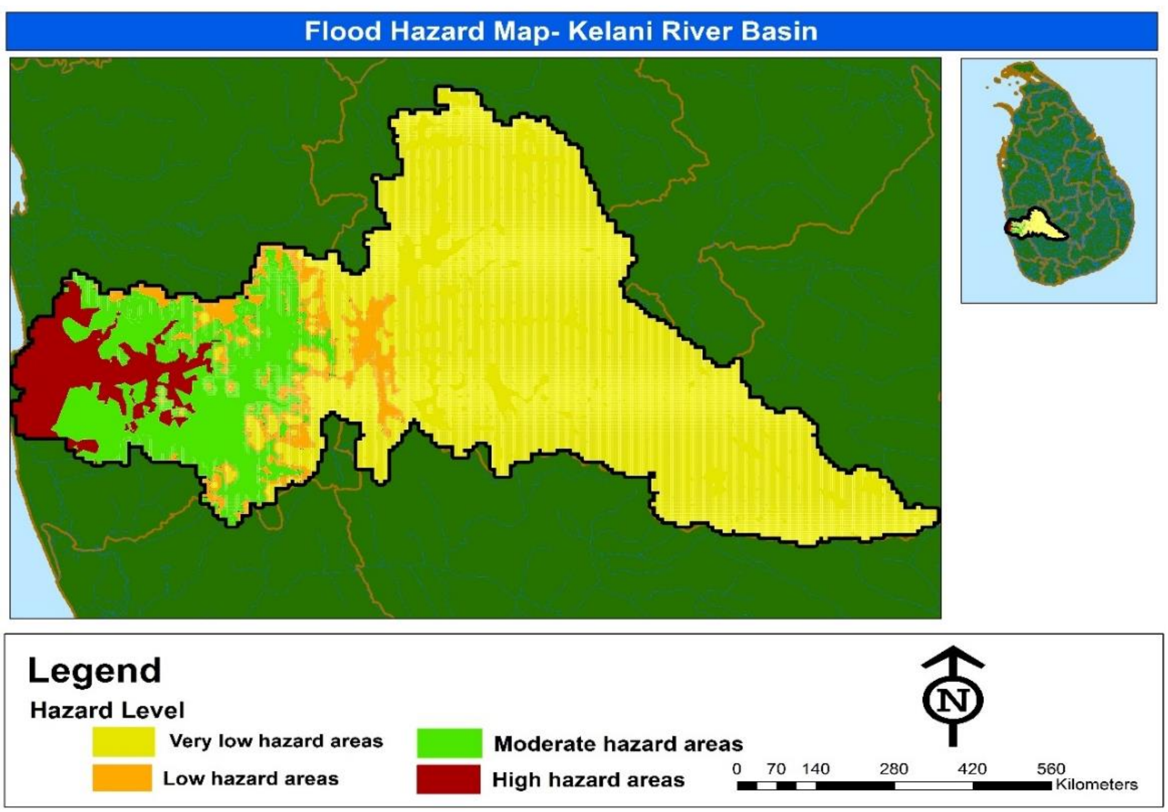

Data Accomplishment
Population Census Data
Flood layer from DEM
Building layers
Identification of vulnerable sub group by GN population
Intersect the flood and GN Boundary
Identification of vulnerable buildings by number of buildings
Cross table
Vulnerability Areas


DEM
Identify the flood level
Flood Hazard Mapping
Vulnerability Mapping
Identify the element at risk
Capacity Analysis
Flood Risk mapping

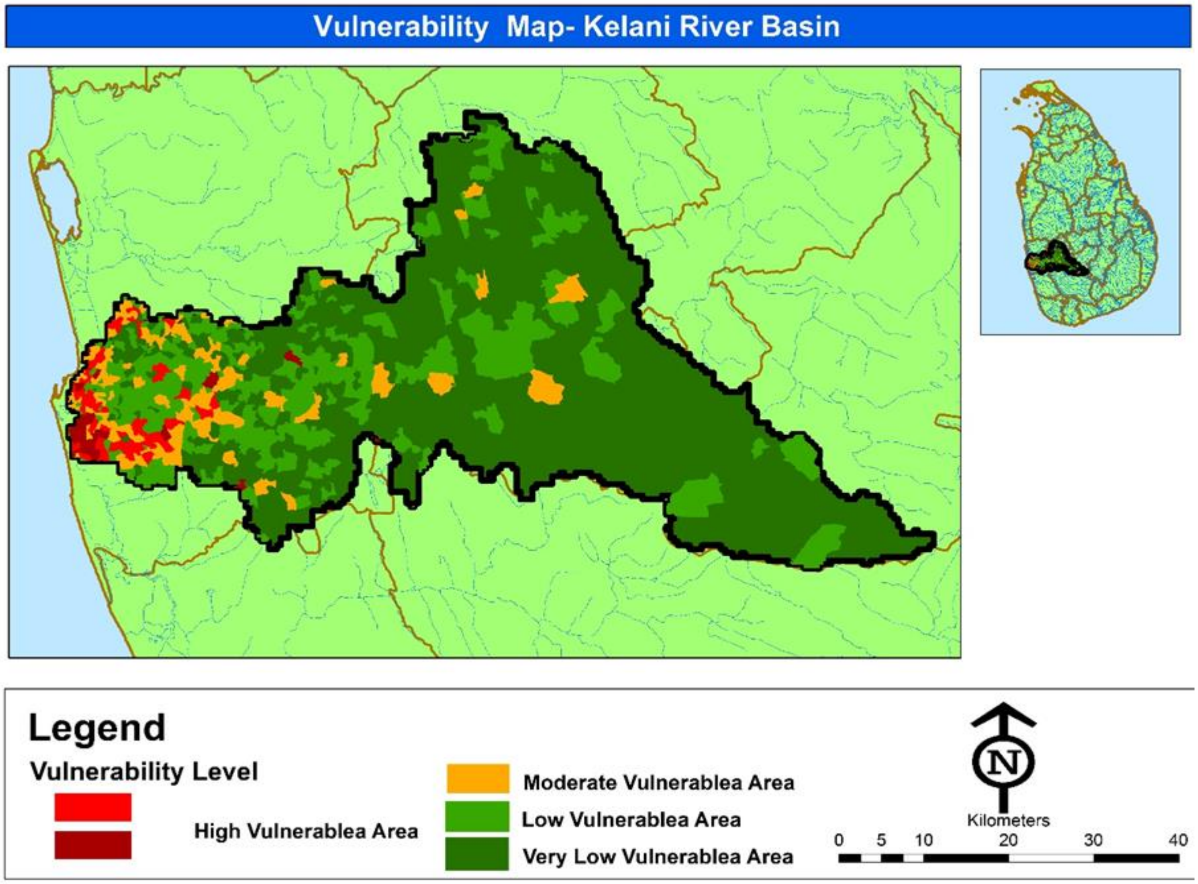
Flood Risk Preparedness Assessment

1. Flood Disaster Map

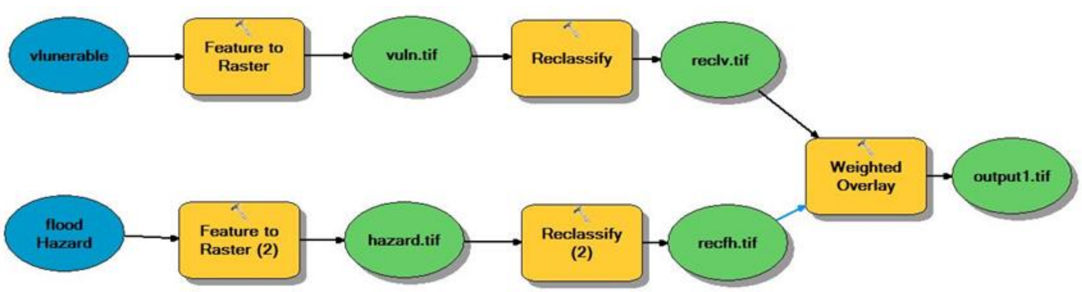


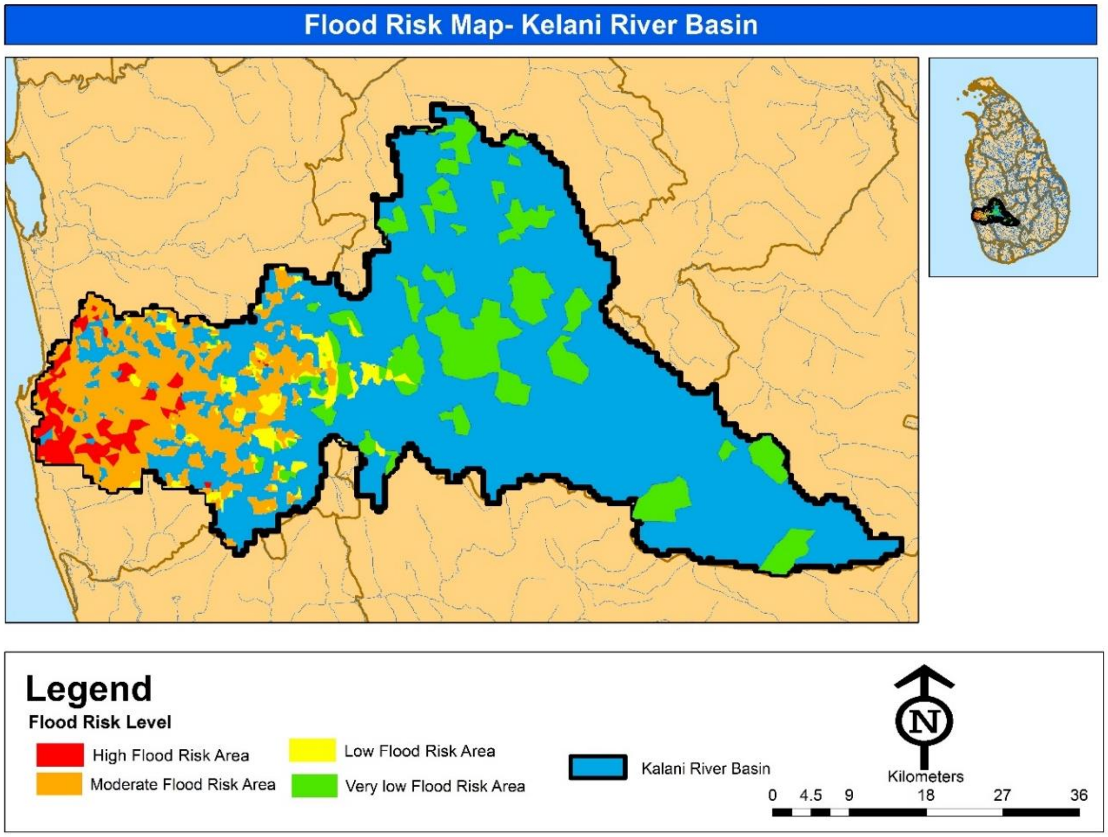


2. Vulnerability analysis of Kelani River area



3. Flood Risk Preparedness Assessment Map





Flood Risk Assessment is helped to identify the flood vulnerability areas. Accordingly, In the future development plans, we could use this information for future development.

Limitations and Potentials

Flood Risk Assessment analyzed under several limitations and potentials.

Potentials

1. GIS technology is an operational tool for monitoring and determining flood risk. With DEM, flood risk assessment factors such as slope, flow direction, and river network can be easily collected.
2. Hazzard mapping methodology is applicable in the local scale. Its advantaged for reduce the disaster and easy to prevent from disaster.
3. Detailed risk maps and trend studies can be constructed using the data available in Sri Lanka. Spatial maps for floods and Flood risks assessment maps are useful for local authorities and international organizations.

Limitations

1. Availability of datas is not enough for vulnerability of mapping.
2. Digitizing exist situation of study area is taken long time.